Research Notes

A Note on the Mixed Polarity of Magnetization in Cainozoic Basalts in Victoria, Australia

W. G. Mumme

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Summary

The mixed polarity of magnetization observed in Cainozoic volcanics is shown, by the use of a.c. demagnetization techniques, to be due to a temporary unstable component of magnetization superimposed on a stable reversed direction.

1. Introduction

Localities of mixed polarity of magnetization were reported by Green & Irving (1958) in their investigation of the palaeomagnetism of the Cainozoic volcanics of Victoria. At Irving's suggestion the author resampled one of these sites, Drouin South, in considerable detail and examined the directions of magnetization of the rocks more fully in order to see if a more complete picture of the mixed polarity could be obtained.

Cylinders of rock were prepared from the samples collected in the usual way, and the directions of magnetization were measured using an astatic magnetometer. The directions of magnetization that were obtained by measurements made at the University of Adelaide are shown in Figure 1. This figure also shows the directions of magnetization determined by Irving and Green. It can readily be seen from this figure that the directions obtained by the author do not fall into two groups 180° apart as was originally thought to be the case, but are much more scattered, indicating the possibility of instability in the magnetization.

2. A.c. demagnetization experiments

The author carried out a.c. demagnetization experiments on these specimens at the Department of Geophysics, A.N.U., Canberra, where a.c. washing apparatus similar to that described by Creer (1958) was made available to him.

By subjecting rock specimens to gradually decreasing alternating magnetic fields it is possible to remove unwanted, unstable components of magnetization which may have been acquired since formation. The a.c. washing in this case was done in peak fields of 150 and 225 oersted which were gradually reduced to zero.

After washing, the directions of magnetization of all except one of the thirteen specimens which were originally normally magnetized became reversed. The initially reversed specimens remained reversed. These directions of magnetization are plotted in Figures 2 and 3, and the values of the precision parameter K (Fisher 1953) for these two distributions are given in Table 1.

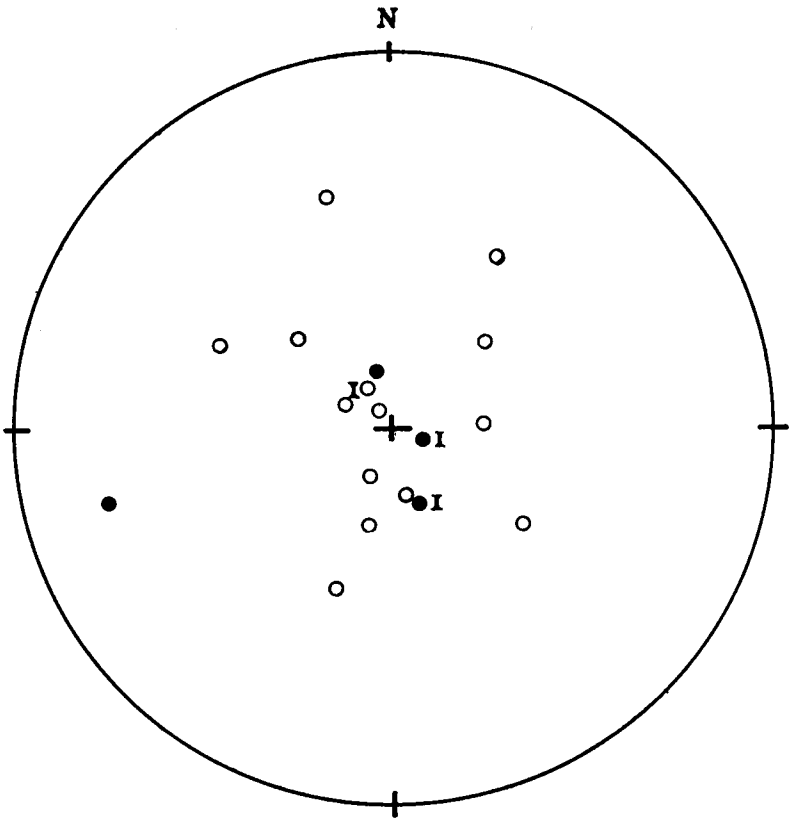


FIG. 1.—Stereographic projection showing original magnetization directions. O point on upper hemisphere. ● point on lower hemisphere. Those points marked I are directions determined by Irving and Green.

Table 1

Washing field	K	Mean direction of magnetization	
		D	I
150	4·4	256	+80
225	3·2	75	+71

Two of the specimens, one originally reversed, the other normally magneuzed were washed progressively in fields of 45, 90, 150, 300, 450, 600, 750 oersted. The a.c. demagnetization curves for these two specimens are shown in Figure 4.

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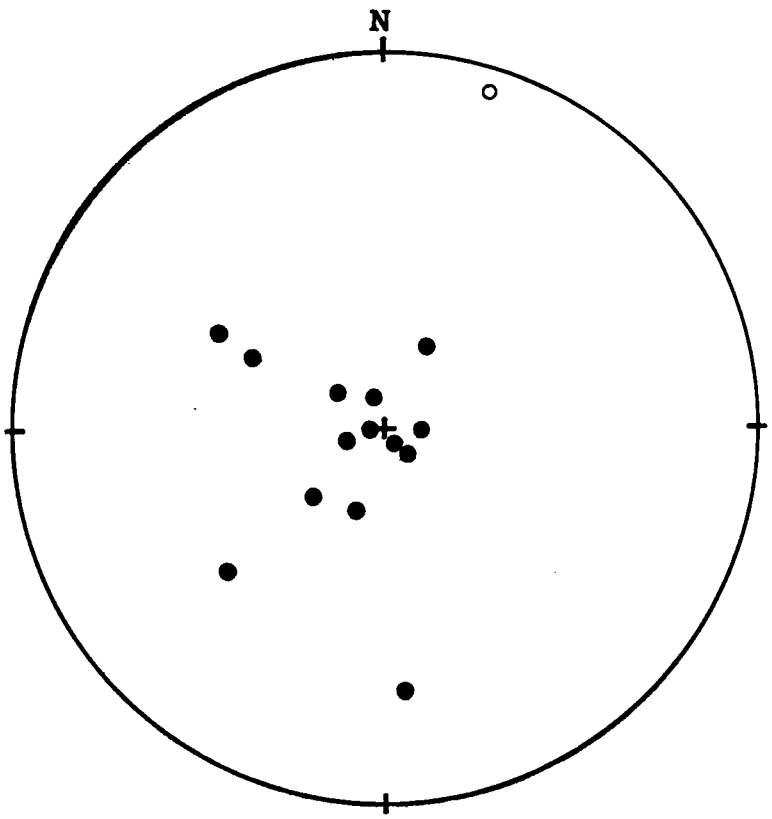


FIG. 2.—Stereographic projection showing directions of magnetization after washing in 150 oersted. O point on upper hemisphere. ● point on lower hemisphere.

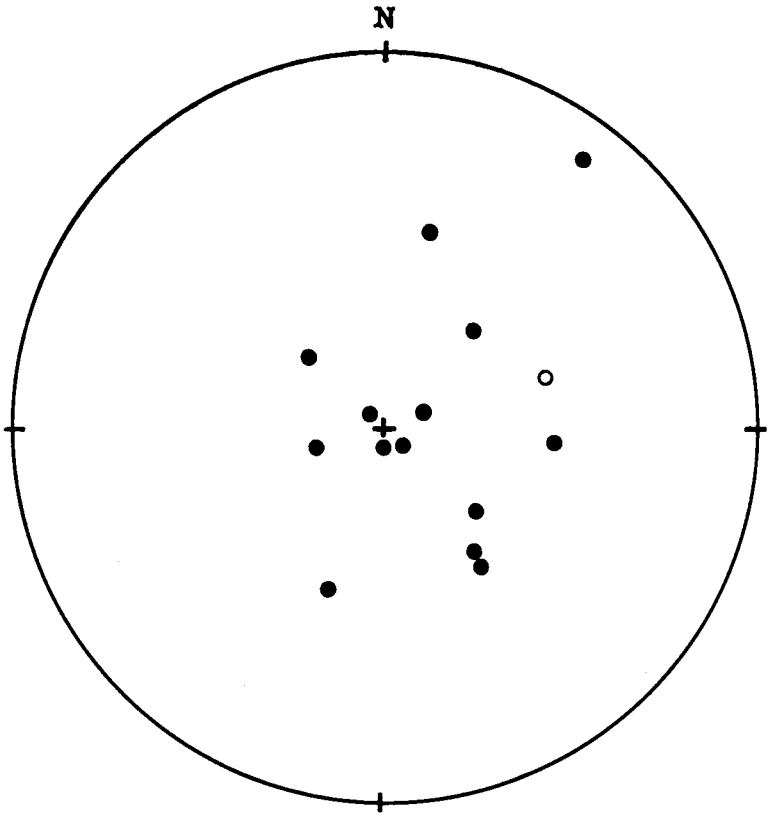


FIG. 3.—Stereographic projection showing directions of magnetization after washing in 225 oersted. O point on upper hemisphere. ● point on lower hemisphere.

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From these two curves it is seen that the magnetization of the reversed specimen is the more stable.

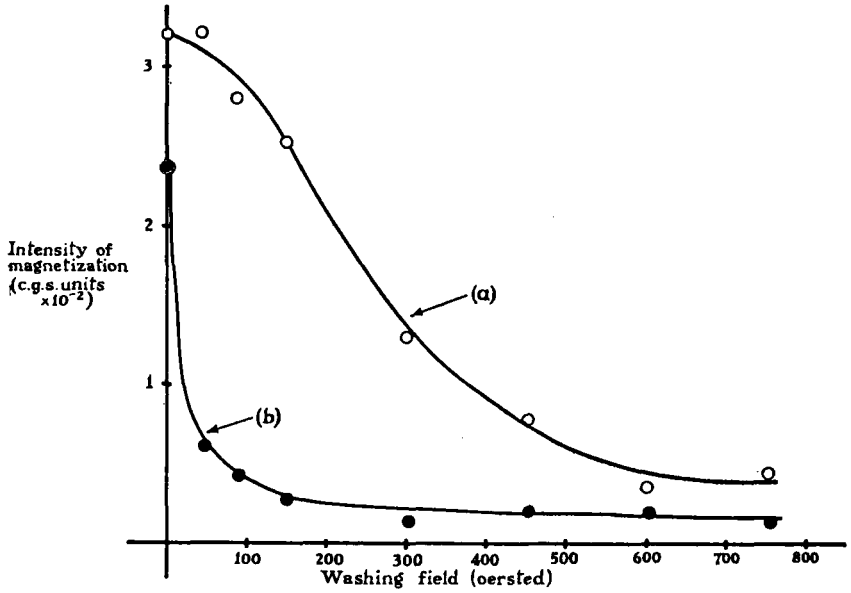


FIG. 4. a.c. Demagnetization curves for: (a) Originally reversed specimen. (b) Originally normally magnetized specimen.

Discussion

The writer carried out this investigation after it was found that the simple picture presented by Green & Irving (1958) of mixed polarities 180° apart existing at Drouin South did not seem to hold when rather more detailed sampling was carried out. It seems clear that the mixed polarities which do exist at Drouin South are caused by a temporary unstable component of magnetization superimposed on a stable reversed direction.

Acknowledgments

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Physics Department,
The University,
Adelaide,
Australia:
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